
EXERCISE

Exercises 2

AI510 - Cybersecurity and Innovation

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1 Exercise 1

1.1 Describes the term “network edge” in your own words.

Endsystems. This is where all applications which we use runs.

- Hosts
 - Clients
 - Servers
 - IoT devices
- Edge devices
 - Routers
 - Modems

1.2 Describes the term “access network” in your own words.

The access network is the entity my data runs through right before it gets to my router.

This could be **fiber cables**, **TV-cables** or **mobile net (like 5G)**

1.3 Describes the term “network core” in your own words.

A network of networks essentially

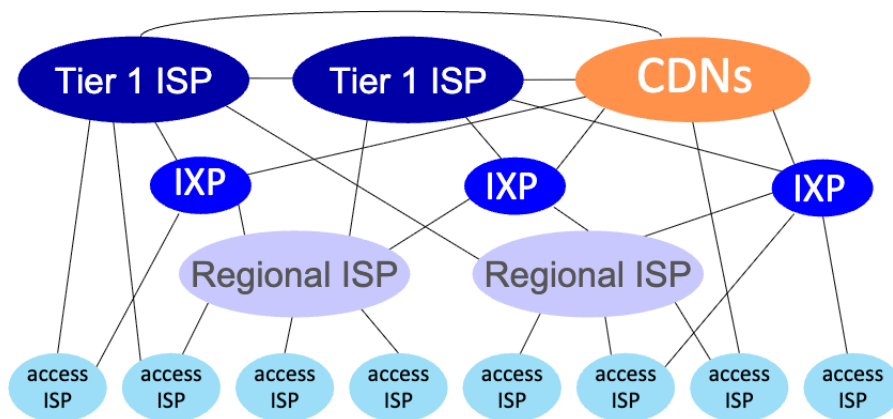


Figure 1: The Internet Topology

The core main function are forwarding and routing

1.4 Which layer is the upper most in the current Internet protocol stack?

Application layer, like HTTP

1.5 Describe the differences between packet and circuit switching

- **Packet switching:** Packet switching allows splits the data dynamically so it only sends data when there is something to send. (When i stop speaking in a call, it stops transmitting)
- **Circuit switching:** “The old way”, allocate the entire line, whenever data is transmitted. (Even if no one is speaking in a call, its still transmitting)

1.6 How are ISPs interconnected?

Via the internet hierarchy

- Tier 1: The Backbone (AT&T or Telia)
- Tier 2: Regional Providers (like a Danish internet provider)
- Tier 3: Local/Access ISP's (More local providers)

1.7 How does packet loss occur?

If the buffer reaches capacity, then additional packets might be lost

1.8 Describe Cerf and Kahn's internetworking principles and how they influence today's Internet in your own words.

- The network-core should be simple. (no need to change the internal network to connect with others)
- Best-effort service model: The network should not promise anything about data-loss. its up to the user to detect if any data is lost.
- Stateless routing: So a router needs as little information about the data as possible (don't need to know what kind of data it is)
- Decentralized control (no master router) every one is responsible for their own router

2 Exercise 2

Assume the network topology as shown below.

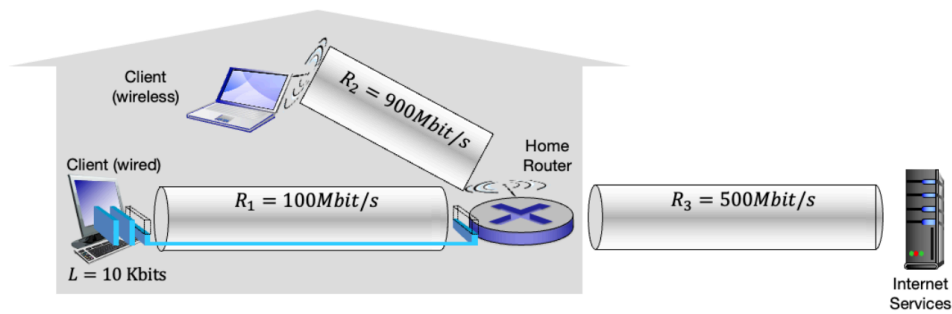


Figure 2:

2.1 What are the bottlenecks in this network for the connection between (a) the wired client and the Internet Services, (b) the wireless client and the Internet Services, (c) the wired and the wireless client?

The smallest link constraints end end throughput

2.2 What is the packet transmission delay for connection R_1 .

$$d_{\text{trans}} = \frac{L}{R_1} = \frac{10 \text{ KBit/s}}{100 \text{ MBit/s}} = \frac{10.000 \text{ Bits}}{100.000.000 \text{ MBit/s}} = 10.000 \text{ s}$$

2.4 How do the bottlenecks for (a), (b), and (c) in 2.1 change if the home router is replaced and R_1 has now a bandwidth of 1000 MBit/s?

$$d_{\text{trans}} = \frac{L}{R_1} = \frac{10 \text{ KBit/s}}{1000 \text{ MBit/s}} = \frac{10.000 \text{ Bits}}{1.000.000.000 \text{ MBit/s}} = 100.000 \text{ s}$$

3 Exercise 3

3.1 For what does a network protocol specify rules?

How to format messages and what actions should be taken when messages are received or other events take place

3.2 Why are network protocols organized in layers?

Such that each part of the protocol is independent, this allows better identification of the data, and eases maintenance

3.3 Explain the term “Encapsulation” in a few sentences

When encapsulation is unique for each layer, its very easily maintained since each layer only cares about 1 type of data.

4 Exercise 4

4.1 Explain the term “Forwarding” in a few sentences.

The local action, while the data is traveling. So at each input throughout the network, the router checks the “plan” made in the routing action.

4.2 Explain the term “Routing” in a few sentences

The initial planning of the best route for the data to travel through the network.

5 Exercise 5

5.1 Explain the term “Packet Sniffing” in a few sentences.

Packets can essentially be picked up by anyone on the same network when it travels. (main defense is encryption)

5.2 Explain the term “IP Spoofing” in a few sentences.

Spoofing is almost the opposite of sniffing, by injecting data which has been tampered with to the receiver

5.3 Explain the term “Denial of Service” in a few sentences. Which categories of the STRIDE model from the Cybersecurity Fundamentals lecture does each of the three terms above correspond?

When an attacker spams a server, overloading it so resources to other become unavailable (DoS)

- Information Disclosure
- Spoofing
- Denial of Service